

Lucas Castillo



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Education

- 2021 – (awaiting viva) **Ph.D. in Psychology** UNIVERSITY OF WARWICK, UK.
Supervisor: Prof. Adam Sanborn
Thesis title: *A Local Sampling Account of Random Generation*
- Jul-Sep 2024 **Research visit** at UNIVERSITY OF HAMBURG, Germany
under the supervision of Prof. Sebastian Gluth.
- 2019 – 2020 **M.Sc. Cognitive and Decision Sciences** UNIVERSITY COLLEGE LONDON, UK
Dissertation title: *Does newly learned causal knowledge guide the perception of temporal order?*
- 2012 – 2016 **B.Sc. Psychology** UNIVERSIDAD DE SEVILLA, Spain.

Acquired Funding

- 2025 **Behaviour Spotlight Small Grant Scheme (£2000)** UNIVERSITY OF WARWICK, UK
(Co-applicant). Project: *Rumination and Random Generation*
- 2024 **Behaviour Spotlight Small Grant Scheme (£2000)** UNIVERSITY OF WARWICK, UK
(Co-applicant). Project: *Exploration Behaviour in Decisions from Experience*

Publications

Journal Articles

- 2 Sanborn, A. N., Zhu, J.-Q., Spicer, J., León-Villagrà, P., **Castillo, L.**, Falbén, J. K., Li, Y.-X., Tee, A., & Chater, N. (2025). Noise in Cognition: Bug or Feature? *Perspectives on Psychological Science*, 20(3), 572–589. <https://doi.org/10.1177/17456916241258951>
- 1 **Castillo, L.**, León-Villagrà, P., Chater, N., & Sanborn, A. N. (2024). Explaining the flaws in human random generation as local sampling with momentum. *PLOS Computational Biology*, 20(1), 1–24. <https://doi.org/10.1371/journal.pcbi.1011739>

Preprints

- 7 **Castillo, L.**, León-Villagrà, P., Falbén, J. K., Chater, N., & Sanborn, A. N. (2025a). *People Need a Few Seconds to Be Random*. PsyArXiv. <https://doi.org/10.31234/osf.io/awg9j>
- 6 **Castillo, L.**, León-Villagrà, P., Falbén, J. K., Chater, N., & Sanborn, A. N. (2025b). *Random Generation is What Comes to Mind*. PsyArXiv. <https://doi.org/10.31234/osf.io/c9xq5>
- 5 **Castillo, L.**, Li, Y.-X., & Sanborn, A. N. (2025). *The samplr package: A tool for modeling human cognition with sampling algorithms*. PsyArXiv. <https://doi.org/10.31234/osf.io/ax8hm>
- 4 Spicer, J., Li, Y.-X., **Castillo, L.**, Falben, J., Qian, S., & Sanborn, A. N. (2025). *Mental Sampling in Preferential Choice: Tracing the Process using Random Generation*. PsyArXiv. <https://doi.org/10.31234/osf.io/kzmb5>

Publications (continued)

- 3 Li, Y.-X., Falben, J., **Castillo, L.**, Spicer, J., Zhu, J.-Q., Qian, S., Chater, N., & Sanborn, A. N. (2025). *Does utility bias mental simulations of risky events?* PsyArXiv. <https://doi.org/10.31234/osf.io/awdr7>
- 2 Falbén, J., **Castillo, L.**, León-Villagrà, P., Chater, N., & Sanborn, A. N. (2024). *Biased mind or biased world? Assessing the accuracy of cultural beliefs that underlie social judgments.* PsyArXiv. <https://doi.org/10.31234/osf.io/5hdbp>
- 1 León-Villagrà, P., **Castillo, L.**, Falbén, J., Chater, N., & Sanborn, A. N. (2024). *Eliciting Beliefs with Random Generation Tasks.* <https://doi.org/10.31234/osf.io/687au>

Peer-reviewed Conference Proceedings Articles

- 4 Valentin, S.[†], **Castillo, L.**[†], Sanborn, A. N., & Lucas, C. G. (2024). Distinguishing Between Process Models of Causal Learning. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46. <https://escholarship.org/uc/item/2p43j2cw>
- 3 Li, Y.-X., Falbén, J. K., **Castillo, L.**, Spicer, J., Zhu, J.-Q., Chater, N., & Sanborn, A. N. (2024). Probability, but not utility, influences repeated mental simulations of lotteries. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 46. <https://escholarship.org/uc/item/2ps2t4sz>
- 2 León-Villagrà, P., **Castillo, L.**, Chater, N., & Sanborn, A. N. (2022). Eliciting Human Beliefs using Random Generation. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 44. <https://escholarship.org/uc/item/88k526dj>
- 1 **Castillo, L.**, León-Villagrà, P., Chater, N., & Sanborn, A. N. (2021). Local Sampling with Momentum Accounts for Human Random Sequence Generation. *Proceedings of the Annual Meeting of the Cognitive Science Society*, 43. <https://escholarship.org/uc/item/3gz154kg>

† = equal contribution

Conferences

Talks

July 2024	MathPsych Characterizing People's Sampling Engines Using Random Generation
August 2023	SPUDM Production rate effects on random generation elicitation
July 2023	MathPsych Modelling speeded random generation as sampling for inference.
July 2022	Virtual MathPsych Using sampling algorithms to explain human random generation.
October 2021	Computational Cognition Workshop Human random generation as a locally-bound process.

Poster Presentations

July 2024	Annual Meeting of the Cognitive Science Society People Need About Five Seconds to be Random: Autocorrelated Sampling Algorithms Can Explain Why
November 2021	Annual Meeting of the Psychonomic Society Local sampling can explain human deviations from randomness.
July 2021	Annual Meeting of the Cognitive Science Society Local sampling with momentum accounts for human random sequence generation.

Reviews

Since	Venue	Count
2024	Journal of Experimental Psychology: General	1
2023	Journal of Experimental Psychology: Learning, Memory, and Cognition	1
2024	Cogsci Conference	10
2023	SPUDM Conference	5

Misc.

Other Outputs

- R Packages **Castillo, L.**, Li, Y.-X., & Sanborn, A. N. (2024). *samplr*: Compare Human Performance to Sampling Algorithms. <https://CRAN.R-project.org/package=samplr>
- Castillo, L.**, Li, Y.-X., & Sanborn, A. N. (2024). *samplrData*: Datasets from the SAMPLING Project. <https://CRAN.R-project.org/package=samplrData>.

Other University Involvement

- 2024 – present **Member of ReproducibiliTea Organizing Committee** at UNIVERSITY OF WARWICK, UK (University-wide Open Research initiative)
- 2022 – 2023 **Student Representative** for Psychology PhD Students at UNIVERSITY OF WARWICK, UK
- 2021 – 2023 **Teaching Assistant**, UNIVERSITY OF WARWICK, UK.
Modules: *Methods and Analysis in Behavioural Science* and
Bayesian Approaches in Behavioural Science
- 2019 – 2020 **Student Representative** for M.Sc. Cognitive and Decision Sciences cohort at UNIVERSITY COLLEGE LONDON, UK

Awards and Scholarships

- 2025 **Best PhD Student Paper** DEPARTMENT OF PSYCHOLOGY, UNIVERSITY OF WARWICK, UK.
Granted for Castillo et al. 2024.
- 2021 **Departmental PhD Scholarship** DEPARTMENT OF PSYCHOLOGY, UNIVERSITY OF WARWICK, UK
- 2020 **Best performing student** of the academic year for the MSc in Cognitive and Decision Sciences, UNIVERSITY COLLEGE LONDON, UK.

Skills

- Languages Spanish (mother tongue), Catalan, English (fluent), French, German (learning!)
- Coding git, Python, R, C++, L^AT_EX, HTML, css, JavaScript, REACTJS.

References

Available upon request